CORE PURE / FURTHER CALCULUS

Integration by Parts

Worked solutions 15 questions · 105 marks

Questions

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1. Use integration by parts to find

$$\int_1^9 \sqrt{x} \ln\left(\frac{9}{x}\right) \mathrm{d}x$$

giving your answer in the form $a + b\ln 3$, where a and b are constants to be found. **[5]**

Solution

Let

$$u = \ln\left(\frac{9}{x}\right), \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \sqrt{x}$$

Then

$$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{x}, \qquad v = \int x^{1/2}\,\mathrm{d}x = \frac{2}{3}x^{3/2}$$

Using integration by parts,

$$\int u \frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v \frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

So if

$$I = \int_1^9 \sqrt{x} \ln\left(\frac{9}{x}\right) \mathrm{d}x$$

then

$$\begin{aligned} I &= \left[\frac{2}{3}x^{3/2}\ln\left(\frac{9}{x}\right)\right]_1^9 - \int_1^9 \frac{2}{3}x^{3/2}\left(-\frac{1}{x}\right)\mathrm{d}x \\ &= \left[\frac{2}{3}x^{3/2}\ln\left(\frac{9}{x}\right)\right]_1^9 + \frac{2}{3}\int_1^9 x^{1/2}\,\mathrm{d}x \end{aligned}$$

Now evaluate the first term:

$$\begin{aligned} \left[\frac{2}{3}x^{3/2}\ln\left(\frac{9}{x}\right)\right]_1^9 &= \frac{2}{3}(9)^{3/2}\ln\left(\frac{9}{9}\right) - \frac{2}{3}(1)^{3/2}\ln\left(\frac{9}{1}\right) \\ &= \frac{2}{3}(27)\ln 1 - \frac{2}{3}\ln 9 \\ &= 0 - \frac{2}{3}(2\ln 3) \\ &= -\frac{4}{3}\ln 3 \end{aligned}$$

Now evaluate the integral:

$$\begin{aligned} \frac{2}{3}\int_1^9 x^{1/2}\,\mathrm{d}x &= \frac{2}{3}\left[\frac{2}{3}x^{3/2}\right]_1^9 \\ &= \frac{4}{9}\left((9)^{3/2} - 1^{3/2}\right) \\ &= \frac{4}{9}(27-1) \\ &= \frac{104}{9} \end{aligned}$$

Hence

$$I = \frac{104}{9} - \frac{4}{3}\ln 3$$

So in the form $a + b\ln 3$,

$$a = \frac{104}{9}, \qquad b = -\frac{4}{3}$$

2. Given that

$$y = \arctan\left(\frac{x-2}{x+2}\right)$$

where $x \neq -2$

(a) find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$, giving your answer in its simplest form. **[2]**

(b) Use integration by parts to find

$$\int \arctan\left(\frac{x-2}{x+2}\right) \mathrm{d}x \qquad \textbf{[4]}$$

Solution

(a) Let

$$u = \frac{x-2}{x+2}$$

so that

$$y = \arctan u$$

Using the chain rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{1+u^2} \cdot \frac{\mathrm{d}u}{\mathrm{d}x}$$

Now differentiate u:

$$\begin{aligned}\frac{\mathrm{d}u}{\mathrm{d}x} &= \frac{(x+2)(1)-(x-2)(1)}{(x+2)^2} \\ &= \frac{x+2-x+2}{(x+2)^2} \\ &= \frac{4}{(x+2)^2}\end{aligned}$$

Hence

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{1}{1+\left(\frac{x-2}{x+2}\right)^2} \cdot \frac{4}{(x+2)^2} \\ &= \frac{4}{(x+2)^2+(x-2)^2} \\ &= \frac{4}{(x^2+4x+4)+(x^2-4x+4)} \\ &= \frac{4}{2x^2+8} \\ &= \frac{2}{x^2+4}\end{aligned}$$

(b) Let

$$I = \int \arctan\left(\frac{x-2}{x+2}\right) \mathrm{d}x$$

Use integration by parts with

$$u = \arctan\left(\frac{x-2}{x+2}\right), \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

Then

$$v = x \qquad \text{and} \qquad \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2}{x^2+4}$$

from part (a).

Using

$$\int u \frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v \frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

gives

$$\begin{aligned} I &= x\arctan\left(\frac{x-2}{x+2}\right) - \int x \cdot \frac{2}{x^2+4}\,\mathrm{d}x \\ &= x\arctan\left(\frac{x-2}{x+2}\right) - \int \frac{2x}{x^2+4}\,\mathrm{d}x \end{aligned}$$

To evaluate the remaining integral, let

$$t = x^2 + 4$$

so that

$$\frac{\mathrm{d}t}{\mathrm{d}x} = 2x \qquad \Rightarrow \qquad \mathrm{d}t = 2x\,\mathrm{d}x$$

Therefore

$$\begin{aligned} \int \frac{2x}{x^2+4}\,\mathrm{d}x &= \int \frac{1}{t}\,\mathrm{d}t \\ &= \ln t + c \\ &= \ln(x^2+4) + c \end{aligned}$$

So

$$I = x\arctan\left(\frac{x-2}{x+2}\right) - \ln(x^2+4) + c$$

Hence,

$$\int \arctan\left(\frac{x-2}{x+2}\right)\mathrm{d}x = x\arctan\left(\frac{x-2}{x+2}\right) - \ln(x^2+4) + c$$

3. Use integration by parts to find

$$\int_0^2 x^2 \ln(1+x)\,\mathrm{d}x$$

giving your answer in the form $a + b\ln 3$, where a and b are constants to be found. **[7]**

Solution

Let

$$I = \int_0^2 x^2 \ln(1+x)\,\mathrm{d}x$$

Use integration by parts with

$$u = \ln(1+x), \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x^2$$

so that

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x}, \qquad v = \frac{x^3}{3}$$

Then

$$\begin{aligned} I &= \left[\frac{x^3}{3}\ln(1+x)\right]_0^2 - \int_0^2 \frac{x^3}{3(1+x)}\,\mathrm{d}x \\ &= \left[\frac{x^3}{3}\ln(1+x)\right]_0^2 - \frac{1}{3}\int_0^2 \frac{x^3}{1+x}\,\mathrm{d}x \end{aligned}$$

Now simplify the fraction:

$$\frac{x^3}{1+x} = x^2 - x + 1 - \frac{1}{1+x}$$

since

$$(1+x)(x^2 - x + 1) = x^3 + 1$$

So

$$\begin{aligned} \int_0^2 \frac{x^3}{1+x}\,\mathrm{d}x &= \int_0^2 \left(x^2 - x + 1 - \frac{1}{1+x}\right)\mathrm{d}x \\ &= \left[\frac{x^3}{3} - \frac{x^2}{2} + x - \ln(1+x)\right]_0^2 \\ &= \left(\frac{8}{3} - 2 + 2 - \ln 3\right) - (0 - 0 + 0 - \ln 1) \\ &= \frac{8}{3} - \ln 3 \end{aligned}$$

Also,

$$\left[\frac{x^3}{3}\ln(1+x)\right]_0^2 = \frac{8}{3}\ln 3$$

Hence

$$\begin{aligned} I &= \frac{8}{3}\ln 3 - \frac{1}{3}\left(\frac{8}{3} - \ln 3\right) \\ &= \frac{8}{3}\ln 3 - \frac{8}{9} + \frac{1}{3}\ln 3 \\ &= 3\ln 3 - \frac{8}{9} \end{aligned}$$

Therefore

$$\int_0^2 x^2 \ln(1+x)\,\mathrm{d}x = -\frac{8}{9} + 3\ln 3$$

So

$$a = -\frac{8}{9}, \qquad b = 3$$

4. Use integration by parts to find

$$\int \operatorname{arsinh}(2x+1)\,\mathrm{d}x$$

giving your answer in terms of logarithms. **[5]**

Solution

Let

$$I = \int \operatorname{arsinh}(2x+1)\,\mathrm{d}x$$

Use the substitution

$$t = 2x+1 \qquad \Rightarrow \qquad \mathrm{d}x = \frac{1}{2}\,\mathrm{d}t$$

Then

$$I = \frac{1}{2}\int \operatorname{arsinh}(t)\,\mathrm{d}t$$

Now integrate by parts. Take

$$U = \operatorname{arsinh}(t), \qquad \frac{\mathrm{d}V}{\mathrm{d}t} = 1$$

so that

$$\frac{\mathrm{d}U}{\mathrm{d}t} = \frac{1}{\sqrt{t^2+1}}, \qquad V = t$$

Using

$$\int U\,\frac{\mathrm{d}V}{\mathrm{d}t}\,\mathrm{d}t = UV - \int V\,\frac{\mathrm{d}U}{\mathrm{d}t}\,\mathrm{d}t$$

gives

$$\int \operatorname{arsinh}(t)\,\mathrm{d}t = t\operatorname{arsinh}(t) - \int \frac{t}{\sqrt{t^2+1}}\,\mathrm{d}t$$

Now

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\sqrt{t^2+1}\right) = \frac{t}{\sqrt{t^2+1}}$$

so

$$\int \frac{t}{\sqrt{t^2+1}}\,\mathrm{d}t = \sqrt{t^2+1}$$

Hence

$$\int \operatorname{arsinh}(t)\,\mathrm{d}t = t\operatorname{arsinh}(t) - \sqrt{t^2+1} + c$$

Therefore

$$I = \frac{1}{2}\left(t\operatorname{arsinh}(t) - \sqrt{t^2+1}\right) + c$$

Substituting back $t = 2x + 1$,

$$I = \frac{1}{2}\Big((2x+1)\operatorname{arsinh}(2x+1) - \sqrt{(2x+1)^2+1}\Big) + c$$

Finally, using

$$\operatorname{arsinh}(y) = \ln\Big(y + \sqrt{y^2+1}\Big)$$

we get

$$\int \operatorname{arsinh}(2x+1)\,\mathrm{d}x = \frac{1}{2}\Big((2x+1)\ln\Big(2x+1+\sqrt{(2x+1)^2+1}\Big) - \sqrt{(2x+1)^2+1}\Big) + c$$

5.

$$h(\theta) = (3\sin\theta - \cos\theta)(\sin\theta + 2\cos\theta)$$

(a) Show that $h(\theta) = \frac{1}{2} - \frac{5}{2}\cos 2\theta + \frac{5}{2}\sin 2\theta$. **[2]**

(b) Hence, using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta\right) h(\theta)\,\mathrm{d}\theta \qquad \textbf{[5]}$$

Solution

(a) Expanding the brackets,

$$\begin{aligned} h(\theta) &= (3\sin\theta - \cos\theta)(\sin\theta + 2\cos\theta) \\ &= 3\sin^2\theta + 6\sin\theta\cos\theta - \sin\theta\cos\theta - 2\cos^2\theta \\ &= 3\sin^2\theta + 5\sin\theta\cos\theta - 2\cos^2\theta \end{aligned}$$

Now use the double-angle identities

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}, \qquad \cos^2\theta = \frac{1+\cos 2\theta}{2}, \qquad \sin\theta\cos\theta = \frac{1}{2}\sin 2\theta$$

Substituting,

$$\begin{aligned} h(\theta) &= 3\left(\frac{1-\cos 2\theta}{2}\right) + 5\left(\frac{1}{2}\sin 2\theta\right) - 2\left(\frac{1+\cos 2\theta}{2}\right) \\ &= \frac{3}{2} - \frac{3}{2}\cos 2\theta + \frac{5}{2}\sin 2\theta - 1 - \cos 2\theta \\ &= \frac{1}{2} - \frac{5}{2}\cos 2\theta + \frac{5}{2}\sin 2\theta \end{aligned}$$

Hence,

$$h(\theta) = \frac{1}{2} - \frac{5}{2}\cos 2\theta + \frac{5}{2}\sin 2\theta$$

(b) Let

$$I = \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta\right) h(\theta)\,\mathrm{d}\theta$$

Using part (a),

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta\right)\left(\frac{1}{2} - \frac{5}{2}\cos 2\theta + \frac{5}{2}\sin 2\theta\right)\mathrm{d}\theta \\ &= \frac{1}{2}\int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta\right)\mathrm{d}\theta - \frac{5}{2}\int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta\right)\cos 2\theta\,\mathrm{d}\theta + \frac{5}{2}\int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta\right)\sin 2\theta\,\mathrm{d}\theta \end{aligned}$$

We evaluate the three integrals separately.

First,

$$\int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta\right) \mathrm{d}\theta = \left[\frac{\pi\theta}{2} - \frac{\theta^2}{2}\right]_0^{\frac{\pi}{2}}$$
$$= \frac{\pi^2}{4} - \frac{\pi^2}{8}$$
$$= \frac{\pi^2}{8}$$

So the first term contributes

$$\frac{1}{2} \cdot \frac{\pi^2}{8} = \frac{\pi^2}{16}$$

Now let

$$B = \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta\right) \cos 2\theta \, \mathrm{d}\theta$$

Integrating by parts with

$$u = \frac{\pi}{2} - \theta, \qquad \frac{\mathrm{d}v}{\mathrm{d}\theta} = \cos 2\theta$$

gives

$$\frac{\mathrm{d}u}{\mathrm{d}\theta} = -1, \qquad v = \frac{1}{2}\sin 2\theta$$

Hence,

$$B = \left[\left(\frac{\pi}{2} - \theta\right)\frac{1}{2}\sin 2\theta\right]_0^{\frac{\pi}{2}} + \frac{1}{2}\int_0^{\frac{\pi}{2}} \sin 2\theta \, \mathrm{d}\theta$$
$$= 0 + \frac{1}{2}\left[-\frac{1}{2}\cos 2\theta\right]_0^{\frac{\pi}{2}}$$
$$= \frac{1}{2}\left(-\frac{1}{2}\cos\pi + \frac{1}{2}\cos 0\right)$$
$$= \frac{1}{2}\left(\frac{1}{2} + \frac{1}{2}\right)$$
$$= \frac{1}{2}$$

Next let

$$C = \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta\right) \sin 2\theta \, \mathrm{d}\theta$$

Again integrating by parts with

$$u = \frac{\pi}{2} - \theta, \qquad \frac{\mathrm{d}v}{\mathrm{d}\theta} = \sin 2\theta$$

gives

$$\frac{\mathrm{d}u}{\mathrm{d}\theta} = -1, \qquad v = -\frac{1}{2}\cos 2\theta$$

So,

$$\begin{aligned} C &= -\left[\left(\frac{\pi}{2}-\theta\right)\frac{1}{2}\cos 2\theta\right]_0^{\frac{\pi}{2}} - \frac{1}{2}\int_0^{\frac{\pi}{2}} \cos 2\theta \,\mathrm{d}\theta \\ &= -\left[\left(\frac{\pi}{2}-\theta\right)\frac{1}{2}\cos 2\theta\right]_0^{\frac{\pi}{2}} - \frac{1}{2}\left[\frac{1}{2}\sin 2\theta\right]_0^{\frac{\pi}{2}} \\ &= \frac{\pi}{4} - 0 \\ &= \frac{\pi}{4} \end{aligned}$$

Substituting these into the expression for I,

$$\begin{aligned} I &= \frac{\pi^2}{16} - \frac{5}{2}\left(\frac{1}{2}\right) + \frac{5}{2}\left(\frac{\pi}{4}\right) \\ &= \frac{\pi^2}{16} - \frac{5}{4} + \frac{5\pi}{8} \\ &= \frac{\pi^2 + 10\pi - 20}{16} \end{aligned}$$

Therefore, the exact value is

$$\int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2}-\theta\right) h(\theta)\,\mathrm{d}\theta = \frac{\pi^2 + 10\pi - 20}{16}$$

This is approximately 1.33.

6. **(a)** Find

$$\int x^2 \arctan x \, \mathrm{d}x \qquad \textbf{[5]}$$

(b) Hence find the exact value of

$$\int_0^1 x^2 \arctan x \, \mathrm{d}x \qquad \textbf{[2]}$$

Solution

(a) Let

$$I = \int x^2 \arctan x \, \mathrm{d}x$$

Use integration by parts with

$$u = \arctan x, \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x^2$$

so that

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2}, \qquad v = \frac{x^3}{3}$$

Then

$$\begin{aligned} I &= uv - \int v \frac{\mathrm{d}u}{\mathrm{d}x} \, \mathrm{d}x \\ &= \frac{x^3}{3} \arctan x - \int \frac{x^3}{3(1+x^2)} \, \mathrm{d}x \\ &= \frac{x^3}{3} \arctan x - \frac{1}{3} \int \frac{x^3}{1+x^2} \, \mathrm{d}x \end{aligned}$$

Now simplify the fraction:

$$\frac{x^3}{1+x^2} = x - \frac{x}{1+x^2}$$

since

$$x - \frac{x}{1+x^2} = \frac{x(1+x^2) - x}{1+x^2} = \frac{x^3}{1+x^2}$$

So

$$\begin{aligned} I &= \frac{x^3}{3} \arctan x - \frac{1}{3} \int \left(x - \frac{x}{1+x^2} \right) \mathrm{d}x \\ &= \frac{x^3}{3} \arctan x - \frac{1}{3} \left(\int x \, \mathrm{d}x - \int \frac{x}{1+x^2} \, \mathrm{d}x \right) \end{aligned}$$

Now

$$\int x \, \mathrm{d}x = \frac{x^2}{2}$$

and, using the substitution $1 + x^2$,

$$\int \frac{x}{1+x^2}\,\mathrm{d}x = \frac{1}{2}\ln(1+x^2)$$

Hence

$$\begin{aligned} I &= \frac{x^3}{3}\arctan x - \frac{1}{3}\left(\frac{x^2}{2} - \frac{1}{2}\ln(1+x^2)\right) + c \\ &= \frac{x^3}{3}\arctan x - \frac{x^2}{6} + \frac{1}{6}\ln(1+x^2) + c \end{aligned}$$

Therefore,

$$\int x^2 \arctan x\,\mathrm{d}x = \frac{x^3}{3}\arctan x - \frac{x^2}{6} + \frac{1}{6}\ln(1+x^2) + c$$

(b) Using the result from part (a),

$$\int_0^1 x^2 \arctan x\,\mathrm{d}x = \left[\frac{x^3}{3}\arctan x - \frac{x^2}{6} + \frac{1}{6}\ln(1+x^2)\right]_0^1$$

Substituting the limits:

$$\begin{aligned} \int_0^1 x^2 \arctan x\,\mathrm{d}x &= \left(\frac{1}{3}\arctan 1 - \frac{1}{6} + \frac{1}{6}\ln 2\right) - \left(0 - 0 + \frac{1}{6}\ln 1\right) \\ &= \frac{1}{3}\cdot\frac{\pi}{4} - \frac{1}{6} + \frac{1}{6}\ln 2 \end{aligned}$$

Since $\ln 1 = 0$,

$$\int_0^1 x^2 \arctan x\,\mathrm{d}x = \frac{\pi}{12} - \frac{1}{6} + \frac{\ln 2}{6}$$

So the exact value is

$$\frac{\pi + 2\ln 2 - 2}{12}$$

7. **(a)** Find

$$\int x^3 \arctan x \, \mathrm{d}x \qquad \textbf{[5]}$$

(b) Hence find the exact value of

$$\int_0^{\sqrt{3}} x^3 \arctan x \, \mathrm{d}x \qquad \textbf{[2]}$$

Solution

(a) Use integration by parts with

$$u = \arctan x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x^3$$

so that

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2}, \qquad v = \frac{x^4}{4}$$

Then

$$\begin{aligned}\int x^3 \arctan x \, \mathrm{d}x &= \int u \frac{\mathrm{d}v}{\mathrm{d}x} \, \mathrm{d}x \\ &= uv - \int v \frac{\mathrm{d}u}{\mathrm{d}x} \, \mathrm{d}x \\ &= \frac{x^4}{4} \arctan x - \int \frac{x^4}{4(1+x^2)} \, \mathrm{d}x \\ &= \frac{x^4}{4} \arctan x - \frac{1}{4} \int \frac{x^4}{1+x^2} \, \mathrm{d}x\end{aligned}$$

Now simplify the fraction by division:

$$\frac{x^4}{1+x^2} = x^2 - 1 + \frac{1}{1+x^2}$$

since

$$(1+x^2)(x^2-1) + 1 = x^4 - 1 + 1 = x^4$$

So

$$\begin{aligned}\int x^3 \arctan x \, \mathrm{d}x &= \frac{x^4}{4} \arctan x - \frac{1}{4} \int \left(x^2 - 1 + \frac{1}{1+x^2} \right) \mathrm{d}x \\ &= \frac{x^4}{4} \arctan x - \frac{1}{4} \left(\int x^2 \, \mathrm{d}x - \int 1 \, \mathrm{d}x + \int \frac{1}{1+x^2} \, \mathrm{d}x \right) \\ &= \frac{x^4}{4} \arctan x - \frac{1}{4} \left(\frac{x^3}{3} - x + \arctan x \right) + c \\ &= \frac{x^4}{4} \arctan x - \frac{x^3}{12} + \frac{x}{4} - \frac{1}{4} \arctan x + c \\ &= \frac{x^4 - 1}{4} \arctan x - \frac{x^3}{12} + \frac{x}{4} + c\end{aligned}$$

Therefore,

$$\int x^3 \arctan x \, \mathrm{d}x = \frac{x^4 - 1}{4} \arctan x - \frac{x^3}{12} + \frac{x}{4} + c$$

(b) Using the result from part (a), let

$$F(x) = \frac{x^4 - 1}{4} \arctan x - \frac{x^3}{12} + \frac{x}{4}$$

Then

$$\int_0^{\sqrt{3}} x^3 \arctan x \, \mathrm{d}x = F(\sqrt{3}) - F(0)$$

First, at $x = \sqrt{3}$,

$$\arctan(\sqrt{3}) = \frac{\pi}{3}, \qquad (\sqrt{3})^4 = 9, \qquad (\sqrt{3})^3 = 3\sqrt{3}$$

so

$$\begin{aligned} F(\sqrt{3}) &= \frac{9-1}{4} \cdot \frac{\pi}{3} - \frac{3\sqrt{3}}{12} + \frac{\sqrt{3}}{4} \\ &= \frac{8}{4} \cdot \frac{\pi}{3} - \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \\ &= \frac{2\pi}{3} \end{aligned}$$

Also,

$$F(0) = \frac{0-1}{4} \arctan 0 - 0 + 0 = 0$$

Hence

$$\int_0^{\sqrt{3}} x^3 \arctan x \, \mathrm{d}x = \frac{2\pi}{3}$$

8. **(a)** Using the logarithmic form of $\operatorname{artanh} x$, prove that the derivative of $\operatorname{artanh} x$ is

$$\frac{1}{1-x^2}$$

for $|x| < 1$. **[3]**

(b) Hence find

$$\int_0^{1/2} \operatorname{artanh} x \, \mathrm{d}x$$

giving your answer in exact logarithmic form. **[5]**

(c) Priya tries to evaluate

$$\int_0^{3/2} \operatorname{artanh} x \, \mathrm{d}x$$

using her calculator, and gets an 'error'. Explain why. **[1]**

Solution

(a) For $|x| < 1$,

$$\operatorname{artanh} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) = \frac{1}{2}\big[\ln(1+x) - \ln(1-x)\big]$$

This is valid for $|x| < 1$ because we need $1 + x > 0$ and $1 - x > 0$.

Differentiate term by term:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\big(\operatorname{artanh} x\big) &= \frac{1}{2}\left[\frac{\mathrm{d}}{\mathrm{d}x}\ln(1+x) - \frac{\mathrm{d}}{\mathrm{d}x}\ln(1-x)\right] \\
&= \frac{1}{2}\left[\frac{1}{1+x} - \frac{1}{1-x}(-1)\right] \\
&= \frac{1}{2}\left[\frac{1}{1+x} + \frac{1}{1-x}\right] \\
&= \frac{1}{2}\left[\frac{(1-x)+(1+x)}{(1+x)(1-x)}\right] \\
&= \frac{1}{2}\left[\frac{2}{1-x^2}\right] \\
&= \frac{1}{1-x^2}
\end{aligned}$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(\operatorname{artanh} x\big) = \frac{1}{1-x^2} \qquad (|x| < 1)$$

(b) Using integration by parts with

$$u = \operatorname{artanh} x, \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

we have

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1-x^2}, \qquad v = x$$

So

$$\begin{aligned}\int \operatorname{artanh} x \,\mathrm{d}x &= x \operatorname{artanh} x - \int x \cdot \frac{1}{1-x^2} \,\mathrm{d}x \\ &= x \operatorname{artanh} x - \int \frac{x}{1-x^2} \,\mathrm{d}x\end{aligned}$$

Now let $t = 1 - x^2$, so

$$\frac{\mathrm{d}t}{\mathrm{d}x} = -2x \qquad \Rightarrow \qquad \mathrm{d}t = -2x \,\mathrm{d}x$$

Then

$$\begin{aligned}\int \frac{x}{1-x^2} \,\mathrm{d}x &= -\frac{1}{2} \int \frac{1}{t} \,\mathrm{d}t \\ &= -\frac{1}{2} \ln t + c \\ &= -\frac{1}{2} \ln(1-x^2) + c\end{aligned}$$

Therefore

$$\int \operatorname{artanh} x \,\mathrm{d}x = x \operatorname{artanh} x + \frac{1}{2} \ln(1-x^2) + c$$

Hence

$$\begin{aligned}\int_0^{1/2} \operatorname{artanh} x \,\mathrm{d}x &= \left[x \operatorname{artanh} x + \frac{1}{2} \ln(1-x^2) \right]_0^{1/2} \\ &= \frac{1}{2} \operatorname{artanh}\left(\frac{1}{2}\right) + \frac{1}{2} \ln\left(\frac{3}{4}\right) - \left(0 + \frac{1}{2} \ln 1\right)\end{aligned}$$

Now

$$\operatorname{artanh}\left(\frac{1}{2}\right) = \frac{1}{2} \ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right) = \frac{1}{2} \ln 3$$

So

$$\begin{aligned}\int_0^{1/2} \operatorname{artanh} x \,\mathrm{d}x &= \frac{1}{2} \cdot \frac{1}{2} \ln 3 + \frac{1}{2} \ln\left(\frac{3}{4}\right) \\ &= \frac{1}{4} \ln 3 + \frac{1}{2}(\ln 3 - \ln 4) \\ &= \frac{1}{4} \ln 3 + \frac{1}{2} \ln 3 - \ln 2 \\ &= \frac{3}{4} \ln 3 - \ln 2\end{aligned}$$

Also,

$$\frac{3}{4} \ln 3 - \ln 2 = \frac{1}{4} \ln\left(\frac{27}{16}\right)$$

Therefore

$$\int_0^{1/2} \operatorname{artanh} x \,\mathrm{d}x = \frac{3}{4} \ln 3 - \ln 2 = \frac{1}{4} \ln\left(\frac{27}{16}\right)$$

(c) In the real number system, $\operatorname{artanh} x$ is defined only for $|x| < 1$. The interval $[0, \frac{3}{2}]$ includes $x = 1$, where $\operatorname{artanh} x$ is undefined, and values $x > 1$, where it is not real. So the calculator gives a domain error.

9. **(a)** Using the logarithmic form

$$\operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right), \qquad |x| < 1$$

prove that the derivative of $\operatorname{artanh} x$ is

$$\frac{1}{1-x^2}$$

[3]

(b) Hence find

$$\int_0^{1/3} \operatorname{artanh}(2x)\,\mathrm{d}x$$

giving your answer in exact logarithmic form. **[5]**

(c) Noah tries to evaluate

$$\int_0^{1/2} \operatorname{artanh}(2x)\,\mathrm{d}x$$

using his calculator, and gets an error. Explain why. **[1]**

Solution

(a) Using the given logarithmic form,

$$\operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) = \frac{1}{2}\big[\ln(1+x) - \ln(1-x)\big]$$

for $|x| < 1$.

Differentiate term by term:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\big(\operatorname{artanh} x\big) &= \frac{1}{2}\left(\frac{\mathrm{d}}{\mathrm{d}x}\ln(1+x) - \frac{\mathrm{d}}{\mathrm{d}x}\ln(1-x)\right) \\
&= \frac{1}{2}\left(\frac{1}{1+x} - \frac{-1}{1-x}\right) \\
&= \frac{1}{2}\left(\frac{1}{1+x} + \frac{1}{1-x}\right) \\
&= \frac{1}{2}\left(\frac{(1-x)+(1+x)}{(1+x)(1-x)}\right) \\
&= \frac{1}{2}\left(\frac{2}{1-x^2}\right) \\
&= \frac{1}{1-x^2}
\end{aligned}$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(\operatorname{artanh} x\big) = \frac{1}{1-x^2}$$

(b) On $0 \le x \le \frac{1}{3}$, we have $0 \le 2x \le \frac{2}{3} < 1$, so $\operatorname{artanh}(2x)$ is defined.

Let

$$I = \int_0^{1/3} \operatorname{artanh}(2x)\,\mathrm{d}x$$

Use integration by parts with

$$u = \operatorname{artanh}(2x), \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

Then

$$v = x$$

and from part (a), using the chain rule,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2}{1-(2x)^2} = \frac{2}{1-4x^2}$$

So

$$\begin{aligned} I &= [x\operatorname{artanh}(2x)]_0^{1/3} - \int_0^{1/3} x \cdot \frac{2}{1-4x^2}\,\mathrm{d}x \\ &= [x\operatorname{artanh}(2x)]_0^{1/3} - \int_0^{1/3} \frac{2x}{1-4x^2}\,\mathrm{d}x \end{aligned}$$

For the remaining integral, let

$$t = 1 - 4x^2$$

Then

$$\mathrm{d}t = -8x\,\mathrm{d}x$$

so

$$2x\,\mathrm{d}x = -\frac{1}{4}\,\mathrm{d}t$$

When $x = 0$, $t = 1$, and when $x = \frac{1}{3}$, $t = \frac{5}{9}$. Therefore

$$\begin{aligned} \int_0^{1/3} \frac{2x}{1-4x^2}\,\mathrm{d}x &= -\frac{1}{4}\int_1^{5/9} \frac{1}{t}\,\mathrm{d}t \\ &= -\frac{1}{4}[\ln t]_1^{5/9} \\ &= -\frac{1}{4}\ln\left(\frac{5}{9}\right) \end{aligned}$$

Hence

$$\begin{aligned} I &= [x\operatorname{artanh}(2x)]_0^{1/3} + \frac{1}{4}\ln\left(\frac{5}{9}\right) \\ &= \frac{1}{3}\operatorname{artanh}\left(\frac{2}{3}\right) + \frac{1}{4}\ln\left(\frac{5}{9}\right) \end{aligned}$$

Now use the logarithmic form again:

$$\begin{aligned}\operatorname{artanh}\left(\frac{2}{3}\right) &= \frac{1}{2}\ln\left(\frac{1+\frac{2}{3}}{1-\frac{2}{3}}\right)\\ &= \frac{1}{2}\ln\left(\frac{5/3}{1/3}\right)\\ &= \frac{1}{2}\ln 5\end{aligned}$$

So

$$\begin{aligned}I &= \frac{1}{3}\cdot\frac{1}{2}\ln 5 + \frac{1}{4}\ln\left(\frac{5}{9}\right)\\ &= \frac{1}{6}\ln 5 + \frac{1}{4}(\ln 5 - \ln 9)\\ &= \frac{1}{6}\ln 5 + \frac{1}{4}\ln 5 - \frac{1}{2}\ln 3\\ &= \frac{5}{12}\ln 5 - \frac{1}{2}\ln 3\end{aligned}$$

This can also be written as

$$\begin{aligned}I &= \frac{1}{12}\left(5\ln 5 - 6\ln 3\right)\\ &= \frac{1}{12}\ln\left(\frac{5^5}{3^6}\right)\\ &= \frac{1}{12}\ln\left(\frac{3125}{729}\right)\end{aligned}$$

Therefore

$$\int_0^{1/3} \operatorname{artanh}(2x)\,\mathrm{d}x = \frac{5}{12}\ln 5 - \frac{1}{2}\ln 3 = \frac{1}{12}\ln\left(\frac{3125}{729}\right)$$

(c) At the upper limit $x = \frac{1}{2}$, we have $2x = 1$, and $\operatorname{artanh}(1)$ is undefined because $\operatorname{artanh} t$ is only defined for $|t| < 1$.

So the integrand is not defined at the endpoint, and a calculator treating this as an ordinary definite integral gives an error.

10. For $0 < x < 1$, use integration by parts to find

$$\int x \operatorname{artanh}(2x-1)\,\mathrm{d}x$$

giving your final answer in terms of logarithms. **[6]**

Solution

Let

$$I = \int x \operatorname{artanh}(2x-1)\,\mathrm{d}x$$

Use integration by parts with

$$u = \operatorname{artanh}(2x-1), \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x$$

Then

$$v = \frac{x^2}{2}$$

and

$$\begin{aligned}\frac{\mathrm{d}u}{\mathrm{d}x} &= \frac{2}{1-(2x-1)^2} \\ &= \frac{2}{1-(4x^2-4x+1)} \\ &= \frac{2}{4x-4x^2} \\ &= \frac{1}{2x(1-x)}\end{aligned}$$

So, by integration by parts,

$$\begin{aligned}I &= uv - \int v\,\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x \\ &= \frac{x^2}{2}\operatorname{artanh}(2x-1) - \int \frac{x^2}{2}\cdot\frac{1}{2x(1-x)}\,\mathrm{d}x \\ &= \frac{x^2}{2}\operatorname{artanh}(2x-1) - \frac{1}{4}\int \frac{x}{1-x}\,\mathrm{d}x\end{aligned}$$

Now simplify the fraction:

$$\frac{x}{1-x} = -1 + \frac{1}{1-x}$$

since

$$-1 + \frac{1}{1-x} = \frac{-(1-x)+1}{1-x} = \frac{x}{1-x}$$

Hence

$$\begin{aligned}\int \frac{x}{1-x}\,\mathrm{d}x &= \int\left(-1+\frac{1}{1-x}\right)\mathrm{d}x \\ &= -x - \ln(1-x)\end{aligned}$$

because

$$\int \frac{1}{1-x}\,\mathrm{d}x = -\ln(1-x)$$

Therefore

$$\begin{aligned} I &= \frac{x^2}{2}\operatorname{artanh}(2x-1) - \frac{1}{4}\big(-x - \ln(1-x)\big) \\ &= \frac{x^2}{2}\operatorname{artanh}(2x-1) + \frac{x}{4} + \frac{1}{4}\ln(1-x) + c \end{aligned}$$

Now write artanh in logarithmic form:

$$\operatorname{artanh} t = \frac{1}{2}\ln\left(\frac{1+t}{1-t}\right)$$

With $t = 2x-1$,

$$\operatorname{artanh}(2x-1) = \frac{1}{2}\ln\left(\frac{1+(2x-1)}{1-(2x-1)}\right) = \frac{1}{2}\ln\left(\frac{2x}{2-2x}\right) = \frac{1}{2}\ln\left(\frac{x}{1-x}\right)$$

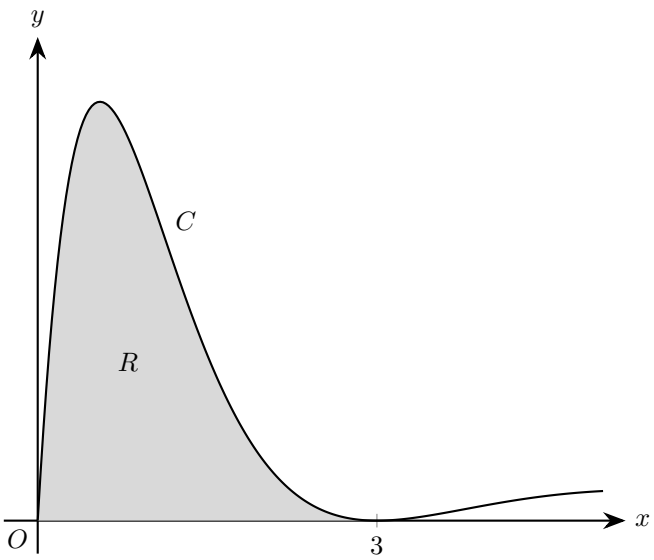
Since $0 < x < 1$, both x and $1-x$ are positive, so these logarithms are valid.

Substitute this into the expression for I:

$$\begin{aligned} I &= \frac{x^2}{2}\cdot\frac{1}{2}\ln\left(\frac{x}{1-x}\right) + \frac{x}{4} + \frac{1}{4}\ln(1-x) + c \\ &= \frac{x^2}{4}(\ln x - \ln(1-x)) + \frac{x}{4} + \frac{1}{4}\ln(1-x) + c \\ &= \frac{x^2}{4}\ln x + \left(-\frac{x^2}{4} + \frac{1}{4}\right)\ln(1-x) + \frac{x}{4} + c \\ &= \frac{x^2}{4}\ln x + \frac{1-x^2}{4}\ln(1-x) + \frac{x}{4} + c \end{aligned}$$

So,

$$\int x\,\operatorname{artanh}(2x-1)\,\mathrm{d}x = \frac{x^2}{4}\ln x + \frac{1-x^2}{4}\ln(1-x) + \frac{x}{4} + c$$



11. The sketch shows part of the curve C given by

$$y = x(3-x)^2e^{-x} \qquad x \geq 0$$

The shaded region R is bounded by C and the x-axis for $0 \leq x \leq 3$.

Find the exact area of R, giving your answer in the form

$$A + Be^{-3}$$

where A and B are rational numbers. **[6]**

Solution

For $0 \leq x \leq 3$,

$$x \geq 0, \qquad (3-x)^2 \geq 0, \qquad e^{-x} > 0$$

so the curve lies above the x-axis, and the required area is

$$\int_0^3 x(3-x)^2e^{-x}\,\mathrm{d}x$$

First expand the bracket:

$$\begin{aligned} x(3-x)^2 &= x(9-6x+x^2) \\ &= 9x - 6x^2 + x^3 \end{aligned}$$

so

$$\text{Area} = \int_0^3 (x^3 - 6x^2 + 9x)e^{-x}\,\mathrm{d}x$$

Now integrate by parts repeatedly, using the tabular method:

Sign	D	I
$+$	$x^3 - 6x^2 + 9x$	e^{-x}
$-$	$3x^2 - 12x + 9$	$-e^{-x}$
$+$	$6x - 12$	e^{-x}
$-$	6	$-e^{-x}$
$+$	0	e^{-x}

Reading the table diagonally gives

$$\begin{aligned}\int (x^3 - 6x^2 + 9x)e^{-x}\,\mathrm{d}x &= -(x^3 - 6x^2 + 9x)e^{-x} - (3x^2 - 12x + 9)e^{-x} \\ &\quad - (6x - 12)e^{-x} - 6e^{-x} + c \\ &= (-x^3 + 3x^2 - 3x - 3)e^{-x} + c\end{aligned}$$

Therefore

$$\begin{aligned}\text{Area} &= \left[(-x^3 + 3x^2 - 3x - 3)e^{-x}\right]_0^3 \\ &= (-27 + 27 - 9 - 3)e^{-3} - \left((-3)e^{0}\right) \\ &= -12e^{-3} + 3 \\ &= 3 - 12e^{-3}\end{aligned}$$

So the exact area is

$$3 - 12e^{-3}$$

which is of the form $A + Be^{-3}$ with

$$A = 3, \qquad B = -12$$

12. Show that

$$\int_{e^{-1/2}}^{1} x(\ln x)^2 \, \mathrm{d}x = a + \frac{b}{e}$$

where a and b are rational constants to be found. **[6]**

Solution

Let

$$I = \int_{e^{-1/2}}^{1} x(\ln x)^2 \, \mathrm{d}x$$

Use the substitution

$$u = \ln x$$

Then

$$x = e^u \qquad \text{and} \qquad \mathrm{d}x = e^u \, \mathrm{d}u$$

So the integrand becomes

$$x(\ln x)^2 \, \mathrm{d}x = e^u \cdot u^2 \cdot e^u \, \mathrm{d}u = u^2 e^{2u} \, \mathrm{d}u$$

Now change the limits.

When $x = e^{-1/2}$,

$$u = \ln\left(e^{-1/2}\right) = -\frac{1}{2}$$

When $x = 1$,

$$u = \ln 1 = 0$$

Hence

$$I = \int_{-1/2}^{0} u^2 e^{2u} \, \mathrm{d}u$$

To integrate this, use integration by parts twice.

First,

$$\int u^2 e^{2u} \, \mathrm{d}u$$

Take

$$p = u^2, \qquad \frac{\mathrm{d}q}{\mathrm{d}u} = e^{2u}$$

Then

$$\frac{\mathrm{d}p}{\mathrm{d}u} = 2u, \qquad q = \frac{1}{2}e^{2u}$$

So

$$\int u^2 e^{2u}\,\mathrm{d}u = \frac{1}{2}u^2 e^{2u} - \int u e^{2u}\,\mathrm{d}u$$

Now integrate $\int ue^{2u}\,\mathrm{d}u$ by parts again.

Take

$$p = u, \qquad \frac{\mathrm{d}q}{\mathrm{d}u} = e^{2u}$$

Then

$$\frac{\mathrm{d}p}{\mathrm{d}u} = 1, \qquad q = \frac{1}{2}e^{2u}$$

Hence

$$\int ue^{2u}\,\mathrm{d}u = \frac{1}{2}ue^{2u} - \frac{1}{2}\int e^{2u}\,\mathrm{d}u = \frac{1}{2}ue^{2u} - \frac{1}{4}e^{2u}$$

Substituting back,

$$\begin{aligned}\int u^2 e^{2u}\,\mathrm{d}u &= \frac{1}{2}u^2 e^{2u} - \left(\frac{1}{2}ue^{2u} - \frac{1}{4}e^{2u}\right)\\ &= e^{2u}\left(\frac{u^2}{2} - \frac{u}{2} + \frac{1}{4}\right)\end{aligned}$$

Therefore

$$I = \left[e^{2u}\left(\frac{u^2}{2} - \frac{u}{2} + \frac{1}{4}\right)\right]_{-1/2}^{0}$$

Now substitute the limits:

$$\begin{aligned}I &= e^0\left(\frac{0^2}{2} - \frac{0}{2} + \frac{1}{4}\right) - e^{-1}\left(\frac{(-1/2)^2}{2} - \frac{-1/2}{2} + \frac{1}{4}\right)\\ &= \frac{1}{4} - \frac{1}{e}\left(\frac{1}{8} + \frac{1}{4} + \frac{1}{4}\right)\\ &= \frac{1}{4} - \frac{1}{e}\left(\frac{5}{8}\right)\\ &= \frac{1}{4} - \frac{5}{8e}\end{aligned}$$

So

$$\int_{e^{-1/2}}^{1} x(\ln x)^2\,\mathrm{d}x = \frac{1}{4} - \frac{5}{8e}$$

Hence it is of the form $a + \frac{b}{e}$ with

$$a = \frac{1}{4}, \qquad b = -\frac{5}{8}$$

13. Given that

$$y = \arcsin\left(\frac{x-1}{3}\right)$$

(a) find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$, giving your answer in its simplest form. **[2]**

(b) Use integration by parts to find

$$\int \arcsin\left(\frac{x-1}{3}\right)\,\mathrm{d}x \qquad \textbf{[4]}$$

Solution

(a) Let

$$u = \frac{x-1}{3}$$

Then

$$y = \arcsin u$$

so by the chain rule,

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{1}{\sqrt{1-u^2}} \cdot \frac{\mathrm{d}u}{\mathrm{d}x} \\ &= \frac{1}{\sqrt{1-\left(\frac{x-1}{3}\right)^2}} \cdot \frac{1}{3} \\ &= \frac{1}{3\sqrt{1-\frac{(x-1)^2}{9}}} \\ &= \frac{1}{\sqrt{9-(x-1)^2}}\end{aligned}$$

(b) Let

$$I = \int \arcsin\left(\frac{x-1}{3}\right)\mathrm{d}x$$

Use integration by parts with

$$u = \arcsin\left(\frac{x-1}{3}\right), \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

Then

$$v = x$$

and from part (a),

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{9-(x-1)^2}}$$

So

$$I = uv - \int v \frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$
$$= x\arcsin\left(\frac{x-1}{3}\right) - \int \frac{x}{\sqrt{9-(x-1)^2}}\,\mathrm{d}x$$

Now write

$$x = (x-1) + 1$$

so

$$\int \frac{x}{\sqrt{9-(x-1)^2}}\,\mathrm{d}x = \int \frac{x-1}{\sqrt{9-(x-1)^2}}\,\mathrm{d}x + \int \frac{1}{\sqrt{9-(x-1)^2}}\,\mathrm{d}x$$

For the first integral, let

$$t = 9-(x-1)^2$$

Then

$$\frac{\mathrm{d}t}{\mathrm{d}x} = -2(x-1) \qquad \Rightarrow \qquad (x-1)\,\mathrm{d}x = -\frac{1}{2}\,\mathrm{d}t$$

Hence

$$\int \frac{x-1}{\sqrt{9-(x-1)^2}}\,\mathrm{d}x = -\frac{1}{2}\int t^{-1/2}\,\mathrm{d}t$$
$$= -t^{1/2}$$
$$= -\sqrt{9-(x-1)^2}$$

For the second integral, using the result from part (a) in reverse,

$$\int \frac{1}{\sqrt{9-(x-1)^2}}\,\mathrm{d}x = \arcsin\left(\frac{x-1}{3}\right)$$

Therefore

$$I = x\arcsin\left(\frac{x-1}{3}\right) - \left(-\sqrt{9-(x-1)^2} + \arcsin\left(\frac{x-1}{3}\right)\right) + c$$
$$= x\arcsin\left(\frac{x-1}{3}\right) + \sqrt{9-(x-1)^2} - \arcsin\left(\frac{x-1}{3}\right) + c$$
$$= (x-1)\arcsin\left(\frac{x-1}{3}\right) + \sqrt{9-(x-1)^2} + c$$

Hence

$$\int \arcsin\left(\frac{x-1}{3}\right)\,\mathrm{d}x = (x-1)\arcsin\left(\frac{x-1}{3}\right) + \sqrt{9-(x-1)^2} + c$$

14.

$$g(\theta) = (1 + \sin\theta - \cos\theta)^2$$

(a) Show that $g(\theta) = 2 + 2\sin\theta - 2\cos\theta - \sin 2\theta$. **[2]**

(b) Hence, using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \theta g(\theta)\,\mathrm{d}\theta \quad \textbf{[5]}$$

Solution

(a) Expanding the square gives

$$\begin{aligned} g(\theta) &= (1 + \sin\theta - \cos\theta)^2 \\ &= 1 + \sin^2\theta + \cos^2\theta + 2\sin\theta - 2\cos\theta - 2\sin\theta\cos\theta \end{aligned}$$

Using $\sin^2\theta + \cos^2\theta = 1$ and $\sin 2\theta = 2\sin\theta\cos\theta$,

$$\begin{aligned} g(\theta) &= 1 + 1 + 2\sin\theta - 2\cos\theta - 2\sin\theta\cos\theta \\ &= 2 + 2\sin\theta - 2\cos\theta - \sin 2\theta \end{aligned}$$

Hence

$$g(\theta) = 2 + 2\sin\theta - 2\cos\theta - \sin 2\theta$$

(b) Let

$$I = \int_0^{\pi/2} \theta g(\theta)\,\mathrm{d}\theta$$

Using part (a),

$$\begin{aligned} I &= \int_0^{\pi/2} \theta(2 + 2\sin\theta - 2\cos\theta - \sin 2\theta)\,\mathrm{d}\theta \\ &= 2\int_0^{\pi/2} \theta\,\mathrm{d}\theta + 2\int_0^{\pi/2} \theta\sin\theta\,\mathrm{d}\theta - 2\int_0^{\pi/2} \theta\cos\theta\,\mathrm{d}\theta - \int_0^{\pi/2} \theta\sin 2\theta\,\mathrm{d}\theta \end{aligned}$$

Now evaluate each integral.

First,

$$\begin{aligned} 2\int_0^{\pi/2} \theta\,\mathrm{d}\theta &= 2\left[\frac{\theta^2}{2}\right]_0^{\pi/2} \\ &= \left[\theta^2\right]_0^{\pi/2} \\ &= \frac{\pi^2}{4} \end{aligned}$$

Next, for $\int_0^{\pi/2} \theta\sin\theta\,\mathrm{d}\theta$, use integration by parts with $u = \theta$, $\frac{\mathrm{d}v}{\mathrm{d}\theta} = \sin\theta$

Then $\frac{\mathrm{d}u}{\mathrm{d}\theta} = 1$ and $v = -\cos\theta$, so

$$\begin{aligned}\int_0^{\pi/2} \theta \sin\theta \,\mathrm{d}\theta &= [-\theta\cos\theta]_0^{\pi/2} + \int_0^{\pi/2} \cos\theta \,\mathrm{d}\theta \\ &= [-\theta\cos\theta + \sin\theta]_0^{\pi/2} \\ &= 1\end{aligned}$$

For $\int_0^{\pi/2} \theta\cos\theta \,\mathrm{d}\theta$, again use integration by parts with $u = \theta$, $\frac{\mathrm{d}v}{\mathrm{d}\theta} = \cos\theta$

Then $\frac{\mathrm{d}u}{\mathrm{d}\theta} = 1$ and $v = \sin\theta$, so

$$\begin{aligned}\int_0^{\pi/2} \theta\cos\theta \,\mathrm{d}\theta &= [\theta\sin\theta]_0^{\pi/2} - \int_0^{\pi/2} \sin\theta \,\mathrm{d}\theta \\ &= [\theta\sin\theta + \cos\theta]_0^{\pi/2} \\ &= \frac{\pi}{2} - 1\end{aligned}$$

For $\int_0^{\pi/2} \theta\sin 2\theta \,\mathrm{d}\theta$, use integration by parts with $u = \theta$, $\frac{\mathrm{d}v}{\mathrm{d}\theta} = \sin 2\theta$

Then $\frac{\mathrm{d}u}{\mathrm{d}\theta} = 1$ and $v = -\frac{1}{2}\cos 2\theta$, so

$$\begin{aligned}\int_0^{\pi/2} \theta\sin 2\theta \,\mathrm{d}\theta &= \left[-\frac{1}{2}\theta\cos 2\theta\right]_0^{\pi/2} + \frac{1}{2}\int_0^{\pi/2} \cos 2\theta \,\mathrm{d}\theta \\ &= \left[-\frac{1}{2}\theta\cos 2\theta + \frac{1}{4}\sin 2\theta\right]_0^{\pi/2} \\ &= \frac{\pi}{4}\end{aligned}$$

Substituting these into I,

$$\begin{aligned}I &= \frac{\pi^2}{4} + 2(1) - 2\left(\frac{\pi}{2} - 1\right) - \frac{\pi}{4} \\ &= \frac{\pi^2}{4} + 2 - \pi + 2 - \frac{\pi}{4} \\ &= \frac{\pi^2}{4} - \frac{5\pi}{4} + 4 \\ &= \frac{\pi^2 - 5\pi + 16}{4}\end{aligned}$$

Therefore,

$$\int_0^{\pi/2} \theta g(\theta) \,\mathrm{d}\theta = \frac{\pi^2 - 5\pi + 16}{4}$$

15. **(i)** Sketch the graph of $y = \arctan x$ for $x \in \mathbb{R}$, indicating any horizontal asymptotes. **[1]**

(ii) Prove that

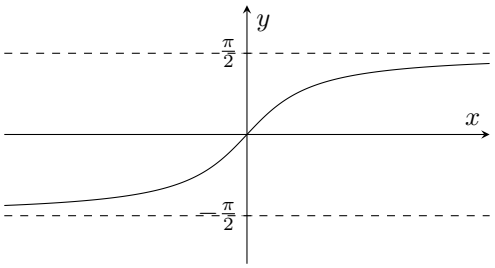
$$\frac{\mathrm{d}}{\mathrm{d}x}(\arctan x) = \frac{1}{1+x^2} \quad \textbf{[3]}$$

(iii) Using integration by parts and suitable substitutions, show that

$$\int_0^1 x(\arctan x)^2 \,\mathrm{d}x = \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2}\ln 2 \quad \textbf{[8]}$$

Solution

(i) The sketch should look like:



The graph of $y = \arctan x$ is an increasing odd curve passing through $(0, 0)$.

As $x \to \infty$, $y \to \frac{\pi}{2}$, and as $x \to -\infty$, $y \to -\frac{\pi}{2}$.

So the horizontal asymptotes are

$$y = \frac{\pi}{2} \qquad \text{and} \qquad y = -\frac{\pi}{2}$$

(ii) Let

$$y = \arctan x$$

Then

$$x = \tan y$$

Differentiate implicitly with respect to x:

$$1 = \sec^2 y \frac{\mathrm{d}y}{\mathrm{d}x}$$

So

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{1}{\sec^2 y} \\ &= \frac{1}{1+\tan^2 y} \\ &= \frac{1}{1+x^2}\end{aligned}$$

since $\tan y = x$.

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}x}(\arctan x) = \frac{1}{1+x^2}$$

(iii) Let

$$I = \int_0^1 x(\arctan x)^2 \,\mathrm{d}x$$

Use integration by parts with

$$u = (\arctan x)^2, \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x$$

Then

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 2(\arctan x)\cdot\frac{1}{1+x^2}, \qquad v = \frac{x^2}{2}$$

Hence

$$\begin{aligned} I &= \left[\frac{x^2}{2}(\arctan x)^2\right]_0^1 - \int_0^1 \frac{x^2}{2}\cdot 2\frac{\arctan x}{1+x^2}\,\mathrm{d}x \\ &= \left[\frac{x^2}{2}(\arctan x)^2\right]_0^1 - \int_0^1 \frac{x^2\arctan x}{1+x^2}\,\mathrm{d}x \\ &= \frac{1}{2}\left(\frac{\pi}{4}\right)^2 - \int_0^1 \frac{x^2\arctan x}{1+x^2}\,\mathrm{d}x \\ &= \frac{\pi^2}{32} - \int_0^1 \frac{x^2\arctan x}{1+x^2}\,\mathrm{d}x \end{aligned}$$

Now

$$\frac{x^2}{1+x^2} = 1 - \frac{1}{1+x^2}$$

so

$$I = \frac{\pi^2}{32} - \int_0^1 \arctan x\,\mathrm{d}x + \int_0^1 \frac{\arctan x}{1+x^2}\,\mathrm{d}x$$

We now evaluate these two integrals separately.

First, let

$$J = \int_0^1 \arctan x\,\mathrm{d}x$$

Using integration by parts with

$$u = \arctan x, \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

gives

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2}, \qquad v = x$$

So

$$J = [x \arctan x]_0^1 - \int_0^1 \frac{x}{1+x^2}\,\mathrm{d}x$$

$$= \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2}\,\mathrm{d}x$$

Now use the substitution $t = 1 + x^2$, so $\mathrm{d}t = 2x\,\mathrm{d}x$. Then

$$\int_0^1 \frac{x}{1+x^2}\,\mathrm{d}x = \frac{1}{2}\int_1^2 \frac{1}{t}\,\mathrm{d}t$$

$$= \frac{1}{2}[\ln t]_1^2$$

$$= \frac{1}{2}\ln 2$$

Therefore

$$J = \frac{\pi}{4} - \frac{1}{2}\ln 2$$

Next, let

$$K = \int_0^1 \frac{\arctan x}{1+x^2}\,\mathrm{d}x$$

Use the substitution

$$t = \arctan x$$

Then

$$x = \tan t, \qquad \mathrm{d}x = \sec^2 t\,\mathrm{d}t, \qquad 1 + x^2 = 1 + \tan^2 t = \sec^2 t$$

When $x = 0$, $t = 0$, and when $x = 1$, $t = \frac{\pi}{4}$. Hence

$$K = \int_0^{\pi/4} \frac{t}{\sec^2 t}\sec^2 t\,\mathrm{d}t$$

$$= \int_0^{\pi/4} t\,\mathrm{d}t$$

$$= \left[\frac{t^2}{2}\right]_0^{\pi/4}$$

$$= \frac{\pi^2}{32}$$

Substitute J and K back into the expression for I:

$$I = \frac{\pi^2}{32} - \left(\frac{\pi}{4} - \frac{1}{2}\ln 2\right) + \frac{\pi^2}{32}$$

$$= \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2}\ln 2$$

Therefore

$$\int_0^1 x(\arctan x)^2\,\mathrm{d}x = \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2}\ln 2$$